

Effectiveness of Immediate Angioplasty or Stenting on Functional Outcomes in Acute Ischemic Stroke With Severe Intracranial Stenosis

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Neurology® 2026;107:e218157. doi:10.1212/WNL.0000000000218157

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Abstract

Background and Objectives

The efficacy of endovascular treatment for acute large vessel occlusion strokes has been demonstrated, but whether it can improve functional outcomes in patients with acute ischemic stroke (AIS) who only present with severe large vessel stenosis without occlusion has not yet been studied. This study investigates the effectiveness of immediate angioplasty or stenting on functional outcomes in AIS patients with severe intracranial stenosis without occlusion.

Methods

We retrospectively included patients with AIS with symptom onset within 24 hours and imaging-confirmed severe intracranial stenosis (70%–99%, Warfarin Aspirin Symptomatic Intracranial Disease criteria) of the culprit vessel from 7 centers in China between January 1, 2020, and December 31, 2024. We compared patients undergoing immediate angioplasty or stenting with those receiving standard medical treatment (SMT) alone. The primary outcome was the distribution of modified Rankin Scale (mRS) scores at 90 days. The treatment effect was estimated through multivariable adjusted models and inverse probability of treatment weighting (IPTW). Safety outcomes included symptomatic intracranial hemorrhage (sICH) within 24 hours and mortality within 90 days.

Results

A total of 242 patients were included, with a mean age of 65.7 years, and 86 patients (35.5%) were female. Ninety-six (39.7%) patients underwent immediate angioplasty or stenting, and 146 (60.3%) received SMT. The median 90-day mRS score was 1 (interquartile range 0–3) in the immediate angioplasty or stenting group and 1 (interquartile range 1–3) in the SMT group. The immediate angioplasty or stenting group showed a shift toward better functional outcomes on the mRS scores (adjusted common odds ratio [OR] 2.73 [95% CI 1.49–5.00], $p = 0.001$; after IPTW, OR 2.50 [95% CI 1.72–3.63], $p < 0.001$). There was no significant difference in the incidence of sICH (1.0% vs 1.4%; adjusted risk ratio 0.36 [95% CI 0.02–5.35], $p = 0.46$) or mortality (2.1% vs 1.4%; adjusted hazard ratio 1.01 [95% CI 0.92–1.11], $p = 0.93$).

Discussion

Among patients with AIS with severe intracranial stenosis without occlusion, immediate angioplasty or stenting was associated with improved 90-day functional outcomes compared with SMT alone. No significant difference was observed in the incidence of sICH or mortality.

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Glossary

AIS = acute ischemic stroke; **ASPECTS** = Alberta Stroke Program Early CT Score; **BAOS** = bailout intracranial angioplasty or stenting; **CTA** = CT angiography; **DSA** = digital subtraction angiography; **END** = early neurologic deterioration; **HR** = hazard ratio; **ICAS** = intracranial atherosclerotic stenosis; **IPTW** = inverse probability of treatment weighting; **IQR** = interquartile range; **IVT** = IV thrombolysis; **LVO** = large vessel occlusion; **MRA** = magnetic resonance angiography; **mRS** = modified Rankin Scale; **NIHSS** = NIH Stroke Scale; **OR** = odds ratio; **PSM** = propensity score matching; **RR** = risk ratio; **sICAS** = symptomatic ICAS; **sICH** = symptomatic intracranial hemorrhage; **SMT** = standard medical treatment.

Classification of Evidence

This study provides Class III evidence that in AIS patients with severe intracranial stenosis, immediate angioplasty or stenting was associated with improved 90-day functional outcomes compared with SMT.

Introduction

Intracranial atherosclerotic stenosis (ICAS) is a leading cause of stroke, with a high prevalence in Asians, accounting for up to 50% of all ischemic strokes.¹ Acute ischemic stroke (AIS) secondary to ICAS may be caused by in situ thrombosis or artery-to-artery embolism, hemodynamic compromise resulting from diminished antegrade flow through high-grade stenosis, and branch atheromatous disease²; therefore, admission imaging may reveal stenosis rather than occlusion in the culprit vessel. Although endovascular treatment is effective for revascularization in large vessel occlusion (LVO) strokes and improves 90-day functional outcomes, it is not recommended in AIS patients with severe intracranial stenosis without occlusion.³ However, persistent severe stenosis of >70% results in continuous cerebral hypoperfusion and hypoxia, as well as increases the risk of recurrent stroke, thereby significantly impeding functional recovery.^{2,4}

Balloon angioplasty or stenting is a practical approach to reversing hemodynamic compromise, improving hypoperfusion, and reducing restenosis in ICAS. Unfortunately, the Stenting vs Aggressive Medical Management for Preventing Recurrent Stroke in Intracranial Stenosis (SAMMPRIS) and Vitesse Intracranial Stent Study for Ischemic Stroke Therapy (VISSIT) trials did not show the superiority of stenting over medical management for symptomatic ICAS (sICAS) within 30 days, with median times from the latest ischemic event to stenting of 7 and 9 days, respectively.^{5,6} The China Angioplasty and Stenting for Symptomatic Intracranial Severe Stenosis (CASSISS) trial excluded sICAS within 3 weeks of symptom onset but also failed to demonstrate the benefit of stenting.⁷ It is suggested that delaying stenting may be unnecessary,⁸ as earlier intervention could effectively stabilize hemodynamics, enhance cerebral perfusion, and preserve ischemic penumbra, leading to better functional outcomes. Notably, AIS patients with ICAS carry a high risk of recurrent stroke and are prone to early neurologic deterioration (END), with the highest END incidence occurring within 24 hours of symptom onset.⁹ Nevertheless, patients with stroke onset within 24 hours, who may be likely to

benefit from revascularization therapy, were excluded from these studies or only accounted for a small subgroup, leaving the effectiveness of immediate angioplasty or stenting for sICAS during the acute phase uncertain.

Therefore, we conducted a multicenter, retrospective, real-world study to assess the effectiveness of immediate angioplasty or stenting on functional outcomes in AIS patients with imaging-confirmed severe intracranial stenosis without occlusion within 24 hours of symptom onset. The primary research question was whether immediate angioplasty or stenting was associated with improved 90-day functional outcomes in AIS patients with severe intracranial stenosis compared with medical treatment.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

The study was approved by the ethics committees of Sun Yat-sen Memorial Hospital, Sun Yat-sen University (SYSKY-2025-116-01), and all participating centers. Informed consent from the patients was waived due to the retrospective design of the study.

Study Design and Patients

This multicenter, retrospective, real-world study included patients from 7 centers in China between January 1, 2020, and December 31, 2024. Clinical data were extracted from the electronic medical records and follow-up databases. The study was conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.¹⁰

Patients were included for the study if they met the following criteria: (1) age of 18 years or older; (2) clinically diagnosed with AIS; (3) time from symptom onset to admission within 24 hours, defined as last-known well time; (4) prestroke modified Rankin Scale (mRS) score of 0–2; (5) Trial of Org 10172 in Acute Stroke Treatment classification of large-artery atherosclerosis; and (6) severe stenosis (70%–99% according

to the Warfarin Aspirin Symptomatic Intracranial Disease method) of the distal intracranial carotid artery, middle cerebral artery (M1 or M2), basilar artery or vertebral artery, diagnosed by digital subtraction angiography (DSA), CT angiography (CTA), or magnetic resonance angiography (MRA) at admission and presumed to be the culprit vessel of the stroke. Exclusion criteria included: (1) occlusion of the culprit vessel, (2) undergoing thrombectomy or intra-arterial thrombolysis, and (3) absence of 90-day follow-up information.

Included patients were categorized into 2 groups based on the treatment they received, including (1) the standard medical treatment (SMT) group, which consisted of patients who received only medical treatment at admission, including IV thrombolysis (IVT), antiplatelet, and anticoagulation treatment; those who received endovascular treatment when progressing to LVO during hospitalization or scheduled endovascular treatment after discharge were also categorized into the SMT group; and (2) the immediate angioplasty or stenting group, which consisted of patients who underwent balloon angioplasty or stenting immediately at admission, including balloon angioplasty alone, balloon-mounted stenting and self-expanding stenting. The medical and endovascular therapeutic strategy was determined at the discretion of the interventionalists based on the patient's clinical status, vascular anatomy, and lesion characteristics, in accordance with each center's protocol.

Data Acquisition

The following baseline and follow-up data were extracted from the electronic medical records, including: (1) demographics: age and sex; (2) medical history: hypertension, diabetes mellitus, hyperlipidemia, coronary heart disease, atrial fibrillation, ischemic stroke, intracranial hemorrhage, cigarette smoking, and alcohol consumption; (3) clinical variables: time from symptom onset to admission; the NIH Stroke Scale (NIHSS) score at baseline, 12–36 hours, and 7 days or discharge, evaluated and documented by the attending neurologists; END, defined as an increase of ≥ 2 points in the total NIHSS score within 7 days after admission⁹; the mRS score at 90 days and any recurrent ischemic events within 90 days, obtained by qualified neurologists and trained staff via clinic visits or telephone interviews at each center; (4) imaging variables: the Alberta Stroke Program Early CT Score (ASPECTS) and perfusion parameters were retrospectively extracted from the baseline neuroimaging reports; the stenosis degree of the culprit vessel was measured by 2 qualified neurologists primarily using DSA as the gold standard, followed by CTA and MRA; symptomatic intracranial hemorrhage (sICH), defined as intracranial hemorrhage assessed by neuroimaging according to the Heidelberg Bleeding Classification that caused new or worsening neurologic symptoms, was ascertained by 2 qualified neurologists through systematic review of medical records and corresponding neuroimaging; (5) treatments: receiving IVT or not according to the European Stroke Organization

guidelines,¹¹ use of tirofiban or not, and the type of endovascular treatments; antiplatelet therapy regimens and other secondary prevention managements within the follow-up period.

Outcome Measurement

The primary outcome was the distribution of 90-day mRS scores. The secondary outcomes included the 90-day mRS score of 0–1 (excellent outcome), 0–2 (functional independence), and 0–3 (favorable outcome), any recurrent ischemic events within 90 days, END, as well as the NIHSS score at 12–36 hours and 7 days (or discharge). Safety outcomes included sICH within 24 hours and mortality within 90 days.

Statistical Analyses

All statistical analyses were conducted using R (version 4.3.1) with a significance threshold of 2-tailed $p < 0.05$. The normality of continuous variables was assessed using the Kolmogorov-Smirnov test. Continuous variables were reported as mean with SD or median with interquartile ranges (IQRs), depending on whether they followed a normal distribution. Categorical variables were presented as counts with percentages. Group comparisons for continuous variables were analyzed using the Student t test or the Mann-Whitney U test, while categorical variables were tested using the χ^2 test.

Missing data in the original data set were addressed using multiple imputation. The imputation model incorporated demographic characteristics, medical history, imaging variables, and treatment to estimate the missing values. A total of 20 imputed data sets were generated. Statistical models were fitted in each imputed training data set, and the results were then pooled to obtain overall estimates of effect sizes and their corresponding standard errors. The primary outcome was analyzed using an ordinal logistic regression model to account for the ordinal nature of mRS scores. The proportional odds assumption of the ordinal logistic regression model was tested and held if $p > 0.05$.¹² In addition, exploratory comparisons of the primary outcome across different endovascular treatment modalities were conducted, with overall significance tested using the likelihood ratio test. Multivariable log-binomial regression models and Cox proportional hazards models were used to examine the binary secondary and safety outcomes. Continuous outcomes were analyzed using quantile regression to model the median, with 95% CIs derived from 500 bootstrap replications. Covariates included in the fully adjusted models were age, sex, baseline NIHSS scores, baseline ASPECTS, IVT, degree of stenosis, time from onset to admission, and use of tirofiban. Multicollinearity and non-linearity were tested. The effect sizes, including common odds ratio (OR), risk ratio (RR), hazard ratio (HR), and median difference, were reported with 95% CIs.

Propensity score-based inverse probability of treatment weighting (IPTW) was used to adjust for confounders and estimate the average treatment effect of immediate

angioplasty or stenting. Variables for weighting methods included age, baseline NIHSS scores, baseline ASPECTS, use of tirofiban, IVT, degree of stenosis, and time from onset to admission. Standardized mean differences across 20 imputed data sets were presented through covariate balance plotting to visualize the balance of baseline characteristics before and after weighting. Weighted analysis was applied to the ordinal logistic regression, log-binomial regression, quantile regression, and Cox proportional hazards models.

Several sensitivity analyses were performed to validate the robustness of the results. First, the impact of missing data was evaluated by excluding samples with missing values. Second, 3 progressively adjusted models were constructed with results pooling from the imputed data sets. Third, propensity score matching (PSM) analysis was conducted in the first imputed data set using a 1:1 ratio and the nearest neighbor method with a caliper of 0.2. Subgroup analyses were performed to determine whether the effectiveness of immediate angioplasty or stenting varied across various subgroups, including sex (male vs female), age (<66 vs ≥ 66 years), baseline NIHSS score (<6 vs ≥ 6), use of tirofiban (yes vs no), IVT (yes vs no), baseline ASPECTS (<8 vs ≥ 8), stenosis location (anterior vs posterior circulation), and degree of stenosis (70%–79% vs 80%–99%).

Data Availability

Data supporting the findings of this study are available from the corresponding author upon reasonable request.

Results

Baseline Characteristics

A total of 768 patients with AIS symptom onset within 24 hours were screened across 7 centers in China between January 1, 2020, and December 30, 2024. After excluding 525 patients who did not meet the criteria at baseline and 1 loss to the 90-day follow-up, 242 patients with imaging-confirmed severe intracranial stenosis without occlusion and complete 90-day outcomes were analyzed. One hundred forty-six patients received SMT alone, and 96 patients underwent immediate angioplasty or stenting. A detailed flow-chart of patient selection is presented in Figure 1. In the immediate angioplasty or stenting group, 28 patients underwent only balloon angioplasty, while 68 patients underwent stenting, including balloon-mounted stent and self-expanding stent.

The overall mean (SD) age was 65.7 (12.3) years, and 156 (64.5%) patients were male. The median (IQR) ASPECTS was 9 (8–10), and the median (IQR) NIHSS scores were 5 (2–8) at baseline. In the immediate angioplasty or stenting group, the median (IQR) degree of stenosis was higher (83% [77%–92%] vs 80% [71%–90%]; $p = 0.04$) and the median (IQR) time from onset to admission was longer (6.6 [3.9–12.1] vs 3.6 [1.9–11.7]; $p = 0.02$). In addition, the

proportion of patients receiving IVT was lower (14.6% vs 38.4%; $p < 0.001$), whereas the proportion receiving tirofiban was higher (75.0% vs 24.7%; $p < 0.001$). Detailed baseline characteristics and follow-up medical management data for all patients and by assigned treatment group are summarized in Table 1 and eTable 1. The results of the normality test are presented in eTable 2 and eFigure 1.

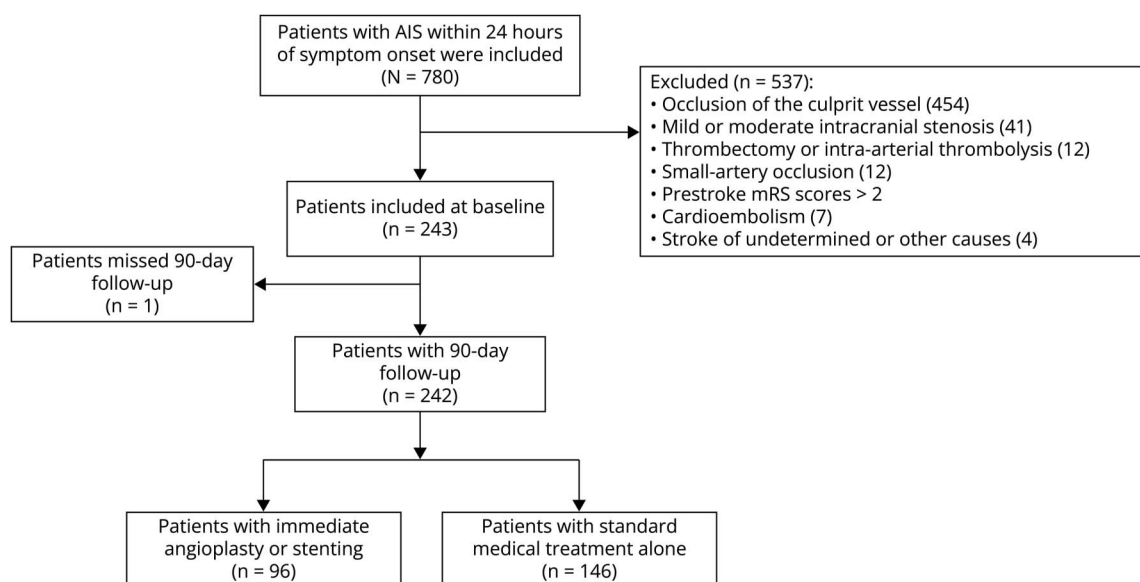
Primary Outcome

The median 90-day mRS score was 1 (IQR 0–3) in the immediate angioplasty or stenting group and 1 (IQR 1–3) in the SMT group. There was an ordinal shift in the 90-day mRS distribution toward a better outcome in angioplasty or stenting compared with SMT in the full adjusted model after pooling the analysis across 20 imputed data sets (adjusted common OR 2.73 [95% CI 1.49–5.00]; $p = 0.001$; Table 2). Proportional odds assumptions held with p for omnibus = 0.34 (eTable 3), and multicollinearity was excluded in all models (eTable 4). Nonlinearity was detected between baseline NIHSS scores and primary outcome (p for nonlinearity = 0.012; eFigure 2); therefore, baseline NIHSS scores were adjusted with a quadratic polynomial term in the ordinal regression model. The results were consistent across each imputed data set (eTable 5). After IPTW, the standardized mean differences for all covariates fell below 0.1, indicating well-balanced characteristics between groups (eFigure 3). Weighted analyses also presented a better treatment effect of immediate angioplasty or stenting over SMT (common OR 2.50 [95% CI 1.72–3.63]; $p < 0.001$; Table 2). The distribution of 90-day mRS scores between the immediate angioplasty or stenting and SMT groups is presented in Figure 2. In addition, models adjusted for perfusion parameters were performed and are presented in eTable 6, yielding consistent results. Exploratory comparisons revealed no significant differences in the primary outcome across endovascular treatment modalities (p for overall = 0.83; eTable 7).

Secondary Outcomes

For secondary outcomes (Table 2), patients receiving immediate angioplasty or stenting were more likely to have excellent functional outcomes (mRS score 0–1; 58.3% vs 50.7%; adjusted RR 1.32 [95% CI 1.08–1.60]; $p = 0.006$) and functional independence (mRS score 0–2; 73.9% vs 70.6%; adjusted RR 1.14 [95% CI 1.00–1.29]; $p = 0.049$) at 90 days than those receiving SMT alone. However, there was no difference in the favorable functional outcome (mRS score 0–3; 85.4% vs 86.3%; adjusted RR 1.05 [95% CI 0.97–1.14]; $p = 0.25$) at 90 days. The immediate angioplasty or stenting group was associated with a lower rate of recurrent ischemic events within 90 days (4.2% vs 14.7%; adjusted RR 0.18 [95% CI 0.06–0.53]; $p = 0.002$). No significant between-group difference in incidence of END was observed (16.8% vs 18.9%; adjusted RR 0.74 [95% CI 0.38–1.44]; $p = 0.37$). The estimated median NIHSS scores were not significantly different between groups at 12–36 hours (median 4 [IQR 2–9] vs 3 [IQR 1–6]; adjusted difference -0.19 [95% CI -1.05 to 0.68];

Figure 1 Flowchart of Selecting AIS Patients With Severe Intracranial Stenosis in This Study



AIS = acute ischemic stroke; mRS = modified Rankin Scale.

$p = 0.67$) and at 7 days or discharge (median 2 [IQR 0–5] vs 1 [IQR 0–4]; adjusted difference -0.36 [95% CI -1.37 to 0.65]; $p = 0.49$). The same results were yielded after IPTW analysis.

Safety Outcomes

Regarding safety outcomes (Table 2), no difference was found in the occurrence of sICH within 24 hours (1.0% vs 1.4%; adjusted RR 0.36 [95% CI 0.02–5.35]; $p = 0.46$) or mortality at 90 days (2.1% vs 1.4%; adjusted HR 1.01 [95% CI 0.92–1.11]; $p = 0.93$) between the 2 groups.

Sensitivity Analysis

The results remained consistent in the fully adjusted model after removing samples with missing values (eTable 8) and in models adjusted for various covariates (eTables 9 and 10). After 1:1 PSM, 57 pairs of patients were matched and balanced (eTable 11, eFigures 4 and 5). The matched patients also exhibited a significant, favorable ordinal shift in 90-day mRS distributions with immediate angioplasty or stenting, as well as a significantly higher proportion of excellent outcomes and functional independence (eTable 12, eFigure 6).

Subgroups Analysis

The trend of treatment effect favoring angioplasty or stenting over SMT remained consistent across subgroups, including sex, age, baseline NIHSS scores, use of tirofiban, IVT, stenosis location, and degree of stenosis (Figure 3). However, there was an interaction between baseline ASPECTS and treatment effect (p for interaction = 0.049). The adjusted common ORs for mRS scores among patients with baseline ASPECTS <8 and ≥ 8 were 1.79 (95% CI 0.43–7.76) and 3.25 (95% CI 1.67–6.44), respectively.

Classification of Evidence

This study provides Class III evidence that in AIS patients with severe intracranial stenosis, immediate angioplasty or stenting was associated with improved 90-day functional outcomes compared with SMT.

Discussion

This multicenter, retrospective, real-world study suggested that in patients with AIS presenting within 24 hours of symptom onset with imaging-confirmed severe intracranial stenosis without occlusion, immediate angioplasty or stenting was associated with an improved 90-day functional outcome compared with SMT alone. No significant difference was observed in the incidence of sICH or mortality. With limited evidence regarding the impact of immediate angioplasty or stenting on functional outcomes in patients with severe large vessel stenosis without occlusion, this study compared the effectiveness and safety of immediate angioplasty or stenting with SMT alone based on contemporary real-world data.

Stroke with underlying ICAS encompasses different pathophysiology, including in situ thrombosis occlusion or artery-to-artery embolism from plaque rupture, hemodynamic compromise, and branch atheromatous disease.² Approximately 30%–40% of occlusion patients may experience vascular recanalization within 24 hours.^{13,14} Therefore, there might be 2 possible underlying mechanisms of our study population, including hemodynamic compromise and partial recanalization of occlusion. Hemodynamic compromise or residual severe stenosis induces significant distal hypoperfusion; meanwhile, hemodynamic compromise may have

Table 1 Baseline Characteristics of Patients

Baseline characteristics ^a	All (N = 242)	Before IPTW		p Value	SMD	After IPTW		SMD
		Immediate angioplasty or stenting (N = 96)	Standard medical treatment (N = 146)			Immediate angioplasty or stenting (N = 189.4)	Standard medical treatment (N = 187.9)	
Age, y, mean (SD)	65.7 (12.3)	65.0 (11.6)	66.1 (12.7)	0.51	0.09	65.5 (11.3)	65.1 (12.6)	0.04
Sex, n (%)								
Male	156 (64.5)	64 (66.7)	92 (63.0)	0.66	0.08	125.3 (66.2)	126.6 (67.4)	0.03
Female	86 (35.5)	32 (33.3)	54 (37.0)			64.1 (33.8)	61.3 (32.6)	
Medical history, n (%)								
Hypertension	159 (65.7)	66 (68.8)	93 (63.7)	0.50	0.11	125.6 (66.3)	117.8 (62.7)	0.08
Diabetes mellitus	76 (31.4)	27 (28.1)	49 (33.6)	0.45	0.12	63.5 (33.5)	70.9 (37.7)	0.09
Hyperlipidemia	101 (41.7)	39 (40.6)	62 (42.5)	0.88	0.04	70.2 (37.1)	67.3 (35.8)	0.03
Coronary heart disease	20 (8.3)	14 (14.6)	6 (4.1)	0.008	0.37	34.5 (18.2)	28.2 (15.0)	0.08
Atrial fibrillation	2 (0.8)	2 (2.1)	0 (0.0)	0.31	0.21	0 (0.0)	0 (0.0)	<0.001
Ischemic stroke	52 (21.5)	19 (19.8)	33 (22.6)	0.72	0.07	35.6 (18.8)	40.5 (21.5)	0.07
Intracranial hemorrhage	3 (1.2)	0 (0.0)	3 (2.1)	0.41	0.21	0 (0.0)	0 (0.0)	<0.001
Cigarette smoking, n (%)	85 (35.1)	32 (33.3)	53 (36.3)	0.74	0.06	55.5 (29.3)	68.9 (36.6)	0.16
Alcohol consumption, n (%)	52 (21.5)	20 (20.8)	32 (21.9)	0.97	0.03	38.1 (20.1)	40.7 (21.7)	0.04
Stenosis sites, n (%)								
Distal intracranial carotid artery	57 (23.6)	26 (27.1)	31 (21.2)	0.50	0.21	57.7 (30.5)	53.3 (28.3)	0.05
Middle cerebral artery	122 (50.4)	46 (47.9)	76 (52.1)			87.5 (46.2)	89.7 (47.8)	
Vertebral artery	39 (16.1)	17 (17.7)	22 (15.1)			29.8 (15.7)	30.8 (16.4)	
Basilar artery	24 (9.9)	7 (7.3)	17 (11.6)			14.4 (7.6)	14.1 (7.5)	
NIHSS score, median (IQR)	5 (2–8)	5.5 (2–11.8)	4 (2–7)	0.12	0.28	4 (2–9)	5 (3–8)	0.03
ASPECTS, median (IQR)	9 (8–10)	9 (8–9)	9 (8–10)	0.14	0.28	9 (8–10)	9 (8–10)	0.01
IV thrombolysis, n (%)	70 (28.9)	14 (14.6)	56 (38.4)	<0.001	0.56	37.4 (19.7)	34.2 (18.2)	0.04
Alteplase	63 (26.0)	11 (11.7)	52 (36.6)			30.6 (16.2)	28.5 (15.2)	
Tenecteplase	6 (2.5)	2 (2.1)	4 (2.8)			3.5 (1.8)	5.7 (3.0)	
Urokinase	1 (0.4)	1 (1.1)	0 (0.0)			3.3 (1.8)	0.0 (0.0)	
Degree of stenosis in %, median (IQR)	80 (73–90)	83 (77–92)	80 (71–90)	0.04	0.31	80 (72–93)	80 (72–92)	0.01
Onset to admission, h, median (IQR)	5.0 (2.2–11.7)	6.6 (3.9–12.1)	3.6 (1.9–11.7)	0.02	0.16	6.0 (2.7–14.2)	7.2 (2.9–13.0)	0.02
Use of tirofiban, n (%)	108 (44.6)	72 (75.0)	36 (24.7)	<0.001	1.17	98.7 (52.1)	100.8 (53.7)	0.03
Type of endovascular treatment, n (%)								
Balloon angioplasty alone	28 (11.6)	28 (29.2)	—	—	—	51.0 (26.9)	—	—
Balloon-mounted stent	15 (6.2)	15 (15.6)	—			37.7 (19.9)	—	
Self-expanding stent	53 (21.9)	53 (55.2)	—			100.7 (53.2)	—	

Abbreviations: ASPECTS = Alberta Stroke Program Early CT Score; IPTW = inverse probability of treatment weighting; IQR = interquartile range; NIHSS = NIH Stroke Scale; SMD = standardized mean difference.

^a Number of missing data for baseline variables: NIHSS score (N = 6), ASPECTS (N = 9), degree of stenosis (N = 80), onset to admission (N = 65). Other variables have no missing data, or the missing data situation has been indicated in the table.

the potential to exacerbate plaque instability due to a heightened mechanical burden.^{15,16} Our study showed that immediate angioplasty or stenting was associated with a lower rate of recurrent ischemic events within 90 days, and with better 90-day functional outcomes. Although no statistical significance was achieved for END, the point estimates consistently favored immediate angioplasty or stenting. Taken together, these findings may indicate that the clinical benefit of immediate angioplasty or stenting may be mediated through its capacity to reverse hemodynamic compromise across the stenosis, rapidly restore blood flow, and maintain luminal patency, potentially reducing the risk of END and recurrent strokes.¹⁷⁻¹⁹

Our findings stood in contrast to the results of major randomized control trials, such as SAMMPRIS, VISSIT, and CASSISS,⁵⁻⁷ which found no superiority of stenting over medical management for sICAS. On the one hand, these trials aimed to evaluate the clinical outcomes of stenting in the setting of secondary stroke prevention, rather than in the setting of acute phase. While SAMMPRIS and VISSIT enrolled patients within 30 days of ischemic events, stenting was performed with a median time from the latest event to stenting of 7 and 9 days, respectively. Notably, neither trial included subgroup analyses of patients treated within 24 hours of symptom onset. Besides, the CASSISS trial may have selected a more stable population by excluding sICAS patients within 3 weeks.²⁰ Actually, AIS patients with ICAS are at high risk of END, with the highest incidence of 43.5% within 24 hours after onset.⁹ Moreover, patients with severe ICAS, particularly those with hemodynamic compromise, remain at high risk of recurrence, with recurrent stroke risks of around 20% at 30 days and up to 37% at 1 year.^{21,22} However, previous trials have overlooked patients with stroke secondary to severe ICAS presented within 24 hours of onset and thus placed interventions well beyond the salvage window. The immediate angioplasty or stenting group in our study received immediate revascularization targeting the hemodynamic compromise and penumbra,²³ which was different from the stable situations addressed in prior trials. Importantly, our study aligns mechanistically with the established benefit of thrombectomy in acute LVO strokes, where time-sensitive reperfusion rescues brain tissue.^{24,25}

On the other hand, advances in neurointerventional techniques and accumulated procedural experience may have increased success in recanalization and mitigated safety issues.^{19,26-28} Beyond the advantage of timely intervention, the superiority of immediate angioplasty or stenting suggested by our study might result from the interventionalists' enhanced ability to select devices tailored to lesion characteristics precisely. Nowadays, the selection of interventions needs full consideration of multiple anatomical factors, including lesion length, vessel diameter, and tortuosity.²⁹ Mori et al.³⁰ classified intracranial stenosis into types A (lesions <5 mm that are concentric or moderately eccentric but not occluded), type B (5–10 mm lesions that are

extremely eccentric or occluded), and type C (lesions >10 mm that are extremely angulated, tortuous, or occluded), with decreasing angioplasty success rates of 92%, 86%, and 33% respectively. In addition, Miao et al.³¹ reported significantly varying technical success rates of tailored interventions based on lesion morphology and vascular access complexity, including 97.5% with balloon mounted stents for smooth access and Mori A lesions, 89.7% with angioplasty for tortuous access and Mori A lesions, and 100% with self-expendable stents for tortuous access and Mori B or C lesions. These studies have deepened the understanding of optimal device-lesion matching and provide evidence for decision making.

Stenting in sICAS was associated with increased vessel perforation and reperfusion injury/hemorrhage in the previous 3 trials. It is evident that angioplasty or stenting for ICAS requires consideration of the risk-benefit balance, as it provides long-term effectiveness but also may result in periprocedural complications.³² Recently, submaximal balloon angioplasty has emerged as a viable therapeutic alternative to mediate this balance, as substantiated by the Balloon Angioplasty for Symptomatic Intracranial Artery Stenosis (BASIS) trial,³³ which has demonstrated the benefits of submaximal balloon angioplasty in reducing periprocedural complications and ensuring long-term efficacy. Since most data in our study were retrospectively collected from medical records, the procedures in the angioplasty-alone subgroup were not predefined, and angioplasty with stenting was predominant in our data. In the BASIS trial, patients in the balloon angioplasty group were enrolled at a median of 34 days after the ischemic event, potentially missing the early high-risk period where END rates peak. Furthermore, risk stratification that combines hemodynamic assessment (e.g., perfusion imaging, fractional flow) and plaque vulnerability assessment (e.g., vessel-wall MRI, fluorodeoxyglucose-PET) is needed to personalize treatment decisions.³² Thus, although BASIS marks a significant advance in treatment strategy, future studies should address timing and risk stratification before widespread clinical implementation.

Our results aligned with previous retrospective studies,³⁴⁻³⁷ which suggested that rescue intracranial angioplasty or stenting following thrombectomy for ICAS-LVO improves 90-day functional outcomes with an acceptable safety profile. However, this perspective was challenged by the results of the Randomized Study of Bailout Intracranial Angioplasty Following Thrombectomy for Acute Large Vessel Occlusion (ANGEL-REBOOT) trial,³⁸ which evaluated the efficacy and safety of bailout intracranial angioplasty or stenting (BAOS) in LVO patients with unsuccessful recanalization or residual stenosis >70% after thrombectomy. ANGEL-REBOOT found no significant difference in 90-day functional outcomes between patients receiving BAOS after thrombectomy and those receiving thrombectomy alone; however, it identified a significantly increased risk of sICH with BAOS. A possible explanation for this discrepancy was that patients in the

Table 2 Outcomes According to Assigned Treatment Group Before and After IPTW

	Immediate angioplasty or stenting (N = 96)	Standard medical treatment (N = 146)	Before IPTW				After IPTW	
			Unadjusted effect size (95% CI)	p Value	Adjusted effect size (95% CI)	p Value	Effect size (95% CI)	p Value
Primary outcome								
mRS score at 90 d, median (IQR) ^a	1 (0–3)	1 (1–3)	1.30 (0.82 to 2.07)	0.26	2.73 (1.49 to 5.00)	0.001	2.50 (1.72 to 3.63)	<0.001
Secondary outcomes								
mRS score of 0–1 at 90 d, n (%) ^b	56 (58.3)	74 (50.7)	1.15 (0.91 to 1.45)	0.24	1.32 (1.08 to 1.60)	0.006	1.54 (1.12 to 2.11)	0.008
mRS score of 0–2 at 90 d, n (%) ^b	71 (73.9)	103 (70.6)	1.05 (0.89 to 1.23)	0.56	1.14 (1.00 to 1.29)	0.049	1.31 (1.05 to 1.65)	0.02
mRS score of 0–3 at 90 d, n (%) ^b	82 (85.4)	126 (86.3)	0.99 (0.89 to 1.10)	0.85	1.05 (0.97 to 1.14)	0.25	1.07 (0.92 to 1.24)	0.36
Any recurrent ischemic events within 90 d ^b	4 (4.2)	21 (14.7)	0.28 (0.10 to 0.80)	0.02	0.18 (0.06 to 0.53)	0.002	0.30 (0.09 to 0.97)	0.047
≥2 points increase in NIHSS score within 7 d after admission ^{b,c}	16 (16.8)	27 (18.9)	0.89 (0.51 to 1.56)	0.69	0.74 (0.38 to 1.44)	0.37	0.72 (0.35 to 1.45)	0.36
NIHSS score, median (IQR) ^d								
At 12–36 h ^e	4 (2–9)	3 (1–6)	1.00 (–1.01 to 3.01)	0.33	–0.19 (–1.05 to 0.68)	0.67	–1.50 (–3.78 to 0.78)	0.20
At 7 d or discharge ^f	2 (0–5)	1 (0–4)	1.00 (–0.28 to 2.28)	0.13	–0.36 (–1.37 to 0.65)	0.49	–1.00 (–2.26 to 0.26)	0.12
Safety outcomes								
Symptomatic intracranial hemorrhage, n (%) ^b	1 (1.0)	2 (1.4)	0.76 (0.07 to 8.27)	0.82	0.36 (0.02 to 5.35)	0.46	0.32 (0.02 to 5.58)	0.44
Death, n (%) ^g	2 (2.1)	2 (1.4)	1.52 (0.21 to 10.83)	0.71	1.01 (0.92 to 1.11)	0.93	1.02 (0.11 to 9.39)	0.99

Abbreviations: IPTW = inverse probability of treatment weighting; IQR = interquartile range; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale.

Adjusted for age, sex, baseline NIHSS scores, baseline ASPECTS, IV thrombolysis, degree of stenosis, onset to admission, and use of tirofiban.

^a Common odds ratio for ordinal logistic regression, with results pooled from 20 multiple imputation data sets. The baseline NIHSS scores were adjusted with a quadratic polynomial.

^b Risk ratios for binary logistic regression, with results pooled from 20 multiple imputation data sets.

^c The early neurologic deterioration was not available for 4 patients due to missing records.

^d Median differences were estimated using quantile regression that models the median (0.5 quantile) with 500 bootstrap replications, and results were pooled from 20 multiple imputation data sets.

^e The NIHSS scores were not available for 9 patients due to missing records.

^f The NIHSS scores were not available for 11 patients due to missing records.

^g Hazard ratio for Cox proportional hazards model, with results pooled from 20 multiple imputation data sets.

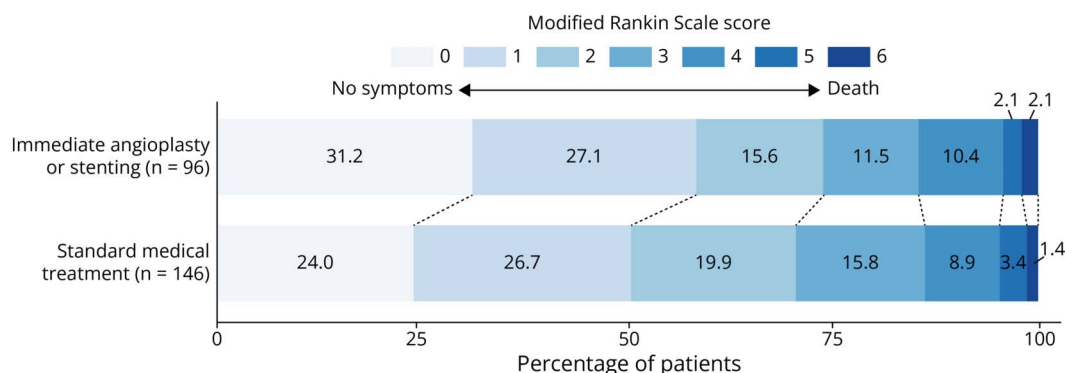
thrombectomy alone group in the trial achieved recanalization through additional thrombectomy attempts and the use of tirofiban.^{39,40} Unlike the ANGEL-REBOOT trial, our study compared direct angioplasty or stenting against medical management alone without additional passes, thereby observing a more pronounced effectiveness difference with a similar incidence of sICH and mortality.

A high ASPECTS has been identified as a predictor of favorable outcomes after endovascular intervention for ischemic stroke.⁴¹ Consistently, our study suggested in subgroup analyses that the association between immediate angioplasty or stenting and clinical outcomes was modified by baseline

ASPECTS, with angioplasty or stenting appearing to confer a more pronounced benefit in patients with baseline ASPECTS ≥8. Further prospective studies are needed to better understand its role in guiding patient selection for immediate angioplasty or stenting.

Several limitations of this study need to be acknowledged. First, the retrospective design is subject to potential selection bias. The data set was not primarily designed to address our study question. Imbalances in baseline characteristics between groups (e.g., time from onset to admission) may introduce bias into the estimated treatment effect. Although we conducted multivariable analyses and sensitivity analyses, the

Figure 2 Distribution of Functional Outcomes at 90 Days in All Patients

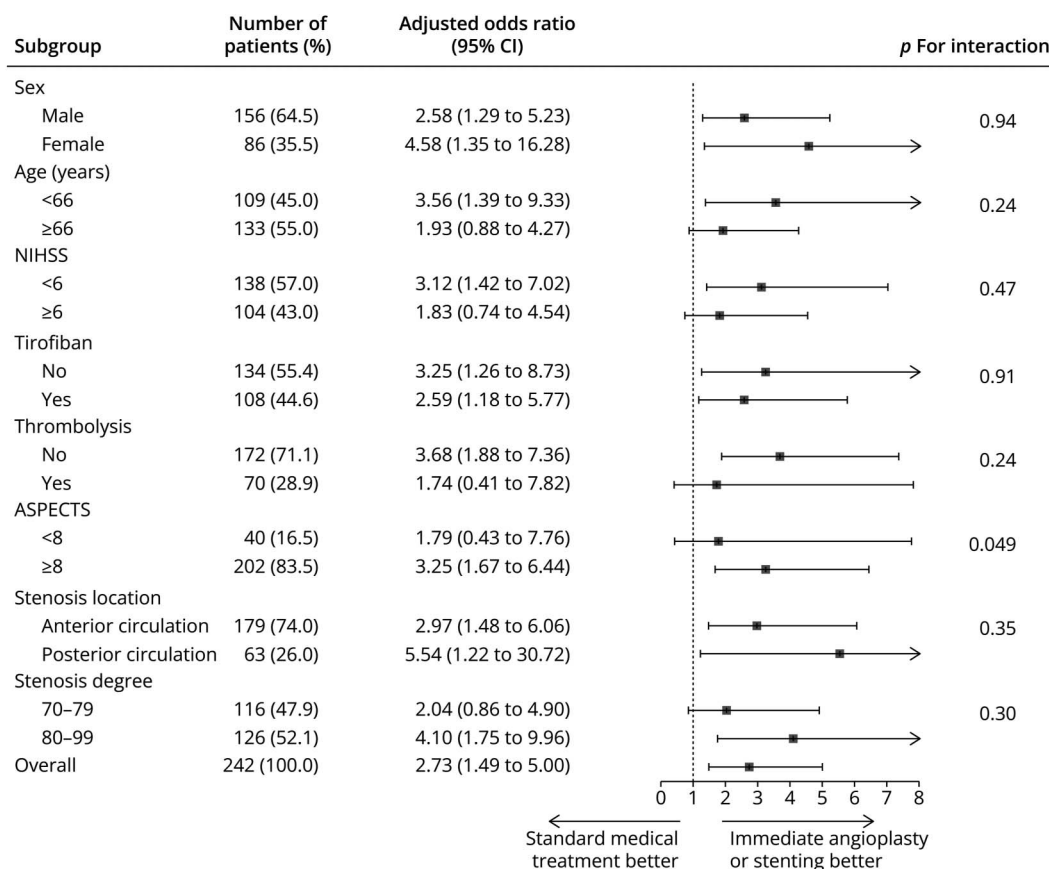


Shown are scores on the modified Rankin Scale for the patients in the 2 treatment groups. Scores range from 0 to 6, with 0 indicating no symptoms, 1 no clinically significant disability, 2 slight disability (patients are able to look after their own affairs without assistance but are unable to carry out all previous activities), 3 moderate disability (patients require some help but are able to walk unassisted), 4 moderately severe disability (patients are unable to attend to bodily needs without assistance and are unable to walk unassisted), 5 severe disability (patients require constant nursing care and attention), and 6 death. The adjusted common odds ratio for the modified Rankin Scale score toward a better outcome with immediate angioplasty or stenting at 90 days was 2.73 (95% CI 1.49–5.00) before IPTW and 2.50 (95% CI 1.72–3.63) after IPTW. IPTW = inverse probability of treatment weighting.

influence of confounding factors cannot be fully eliminated. Consequently, the reported statistical significance should be interpreted with caution. Second, key variables, including

perfusion parameters, vessel tortuosity, and stenosis length, were incomplete or absent. Third, the study lacked standardization in the intervention procedures, such as the times

Figure 3 Subgroup Analyses of the Primary Outcome



Subgroup analyses of the primary outcome by ordinal logistic regression. The forest plot shows the effects of immediate angioplasty or stenting on the primary outcome (common odds ratios indicating the addition of 1 point on the modified Rankin scale) at 90 days across various subgroups. Adjusted confounders included age, sex, baseline NIHSS scores with a quadratic polynomial term, baseline ASPECTS, IV thrombolysis, degree of stenosis, onset to admission, and use of tirofiban. ASPECTS = Alberta Stroke Program Early CT Score; NIHSS = NIH Stroke Scale.

of balloon inflations and the types of self-expanding stents. Fourth, operator and center effects stemming from differing levels of experience across centers constitute an additional limitation. Future prospective multicenter randomized trials with standardized procedures, systematic assessment of hypoperfusion, and inclusion of hospitals with varying levels of expertise are warranted to validate our findings. Importantly, a clinical trial (NCT06437600) is currently underway to evaluate the efficacy and safety of immediate angioplasty for AIS with severe ICAS.

In conclusion, this multicenter retrospective real-world study suggested that in patients with AIS with onset within 24 hours and imaging-confirmed severe intracranial stenosis without occlusion, immediate angioplasty or stenting was associated with an improved 90-day functional outcome compared with SMT alone. There was no significant difference in the incidence of sICH within 24 hours or in mortality within 90 days. Our findings require validation through further prospective, multicenter, randomized trials.

Acknowledgment

The authors thank the staff involved in data collection and patients' follow-up.

Author Contributions

X. Chen: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. T. Yi: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. W. Cao: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. Y. He: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. H. Sun: major role in the acquisition of data; analysis or interpretation of data. Z. Deng: major role in the acquisition of data; analysis or interpretation of data. Y. Chen: major role in the acquisition of data; analysis or interpretation of data. B. He: major role in the acquisition of data. Y.T. Xu: major role in the acquisition of data. Y. Zhu: analysis or interpretation of data. R. Xue: analysis or interpretation of data. X. Yang: analysis or interpretation of data. D. Li: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. Y. Tang: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. L. Zhou: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

Study Funding

This work was supported by grants from STI 2030 Major Projects (2022ZD0211603), National Natural Science Foundation of China (82330099), Science and Technology Program of Guangzhou (2023A03J0708), Guangdong S&T Program

(2023B0303040003), National Natural Science Foundation of China (82371369), Sun Yat-sen Memorial Hospital Clinical Research 5010 Program (SYS-5010-202402), Guangzhou Science and Technology Programme (2024A03J0911), National Natural Science Foundation of China (82473563), Sun Yat-sen Pilot Scientific Research Fund (SYSQH-II-2024-04), and National Natural Science Foundation of China (82304120).

Disclosure

The authors report no relevant disclosures. Go to [Neurology.org/N](https://www.neurology.org/N) for full disclosures.

Publication History

Received by *Neurology*® October 14, 2025. Accepted in final form April 6, 2026. Submitted and externally peer reviewed. The handling editor was Editor-in-Chief José Merino, MD, MPhil, FAAN.

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